

QUANTUM COMPUTING & AI IN QUANTUM

INDUSTRY RESEARCH REPORT

Segment 1 of 3: Tech Giants

Problem • Solution • Products • Big Wins • Best Practices • Anti-Patterns

IBM

GOOGLE

MICROSOFT

AWS

NVIDIA

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SECTION 1

Executive Summary

2025-2026 marked the shift from 'quantum theory' to 'quantum pilots in production.' McKinsey's 2026 Quantum Technology Monitor found that a third of large enterprises now allocate more than \$10 million annually to quantum initiatives, with quantum computing companies collectively crossing \$1 billion in revenue in 2025 — a figure expected to reach \$4.4 billion by 2028.

Every major tech giant has converged on the same architectural pattern: hybrid quantum-classical pipelines, accessed via cloud APIs, targeting near-term advantage in chemistry simulation, optimisation, and quantum-enhanced machine learning. The differentiation is in hardware modality, software ecosystem maturity, and how aggressively each player frames quantum timelines.

THE CORE PROBLEM EVERY GIANT IS SOLVING

Enterprises sit on classically-intractable problems — molecular simulation for drug discovery, portfolio optimisation across thousands of assets, supply chain routing with combinatorial explosion, and machine learning models that hit diminishing returns from scaling alone. Classical compute, even at exascale, cannot efficiently represent the exponential state spaces these problems require. The shared bet: quantum processors, even noisy near-term ones, can shortcut specific sub-routines within these larger workflows when paired correctly with classical AI.

SECTION 2

IBM — Enterprise Quantum & Fault-Tolerance Roadmap

IBM Quantum

280+ ecosystem partners • Path to error-corrected QC by 2028

THE PROBLEM

Enterprises in regulated industries (banking, pharma, logistics) need a credible, auditable path from 'quantum experiment' to 'quantum production workload' — without re-architecting their entire technology stack or betting on a single unproven hardware modality.

THE SOLUTION / APPROACH

IBM combined the largest installed base of cloud-accessible superconducting QPUs (Eagle, and the newly launched Nighthawk processor) with Qiskit — the most widely adopted quantum SDK — and a structured roadmap toward fault-tolerant computing by 2028. IBM Consulting embeds quantum specialists directly into client transformation programmes, turning quantum from an R&D; curiosity into a budget line within existing IT modernisation projects.

KEY PRODUCTS & PLATFORMS

- Qiskit (SDK) + Qiskit Runtime v2 (Sampler/Estimator primitives, Sessions)
- IBM Quantum Network — 20-30 cloud-accessible QPUs (Eagle, Nighthawk processors)
- IBM Quantum Network — enterprise partnership programme (280+ organisations)
- Qiskit Functions Catalog — pre-packaged algorithm-as-a-service modules
- Project Bob — AI-powered software development; 8,000+ internal devs, 45% productivity gain

FLAGSHIP USE CASES & RESULTS

Use Case	Domain	Result / Metric
Quantum chemistry simulation	Pharma / Materials	Real-use-case milestone: 'industry's best-available quantum computers' now used for actual drug development and materials problems per IBM Research
Portfolio & risk optimisation	Financial Services	Embedded via IBM Consulting engagements with Morgan Stanley, Prudential and other Quantum Summit speakers
AI-augmented software delivery	Cross-industry	8,000+ internal devs using Project Bob, averaging 45% productivity gains — 'Client Zero' proof point
Quantum networking	Telecom Infrastructure	/ Cisco partnership on quantum networking announced 2025

BIG WINS

IBM's stock climbed ~27% in 2025 (market cap ~\$260B), with quantum cited as a credible long-term growth lever alongside AI and hybrid cloud.

Outlined the most detailed fault-tolerance roadmap in the industry — explicit milestones toward error-corrected quantum computing by 2028, giving enterprise planners a concrete timeline to build against.

Quantum Summit attendance nearly doubled year-over-year, with Fortune 500 speakers (United Airlines, T-Mobile, UPS, Verizon, Cigna) presenting live use cases rather than theoretical interest.

BEST PRACTICES

- ✓ Treat quantum as a long-horizon R&D; line item embedded within an existing digital transformation budget — not a standalone moonshot.
- ✓ Use Qiskit Functions / pre-packaged algorithm modules to abstract circuit-level complexity from domain teams.
- ✓ Pair every quantum pilot with a named IBM Consulting or partner team that has run the algorithm class before — chemistry, optimisation, and ML each require different expertise.
- ✓ Build internal capability via the IBM Quantum Network rather than one-off vendor engagements — the network model creates compounding institutional knowledge.

ANTI-PATTERNS TO AVOID

- ✗ Expecting quantum advantage on general-purpose ML workloads today — IBM's own framing is 'near-term use cases' in chemistry and optimisation, not broad AI replacement.
- ✗ Running pilots as isolated science projects disconnected from a production data pipeline — IBM's most cited wins are tied to existing enterprise workflows (Project Bob inside SDLC, portfolio tools inside risk systems).
- ✗ Choosing optimization_level=0 / minimal transpilation settings for production-bound code, then being surprised by poor fidelity on real hardware.
- ✗ Ignoring the 280-partner ecosystem and attempting to build quantum chemistry expertise from scratch in-house — this dramatically extends time-to-value.

Sources: IBM 3Q25 Earnings (SEC 8-K); IBM Think 'Trends Shaping AI and Tech in 2026'; Constellation Research 2025 Year in Review; LevelFields Quantum Stocks 2026.

SECTION 3

Google Quantum AI — Willow & Algorithmic Breakthroughs

Google Quantum AI

Willow chip • 13,000x simulation speedup demonstrated Oct 2025

THE PROBLEM

Physics, chemistry, and materials science simulations that scale exponentially with system size (number of interacting particles, electrons, or atoms) are fundamentally beyond reach of classical supercomputers for problems of real-world relevance — even with the largest HPC clusters available.

THE SOLUTION / APPROACH

Google focused its public narrative on demonstrable, verifiable quantum advantage on specific physics simulation tasks using the Willow chip, while keeping commercial access more limited and partnership-driven than IBM or AWS. The strategy: prove the 'why quantum matters' case with headline-grabbing benchmarks, then convert that credibility into research partnerships and eventual Google Cloud integration.

KEY PRODUCTS & PLATFORMS

- Willow quantum chip — 105 qubits, focused on error-correction research
- Cirq SDK — Python framework for quantum circuits, research-oriented
- Google Quantum AI research partnerships (limited public cloud access vs IBM/AWS)
- TensorFlow Quantum — bridges quantum circuits with TensorFlow ML pipelines

FLAGSHIP USE CASES & RESULTS

Use Case	Domain	Result / Metric
Physics simulation benchmark	Research / Materials	13,000x speedup over Frontier supercomputer using 65 qubits (Oct 2025) for a real physics simulation task
Quantum error correction research	Hardware R&D;	Landmark milestones in below-threshold error correction — directly informs IBM/AWS/Microsoft roadmaps as a shared reference point
Quantum-enhanced chemistry	Materials Science	Willow chips actively used for AI/chemistry simulation work per industry tracking (SpinQ 2025 footprint report)

BIG WINS

October 2025: demonstrated a 13,000x speedup over the Frontier supercomputer using only 65 qubits for a physics simulation — one of the most concrete, verifiable quantum advantage claims to date.

Willow's error-correction results set a new public benchmark that competitors (IBM, Microsoft, IonQ) now reference in their own roadmap communications — Google effectively set the industry's technical bar.

Despite NVIDIA's CEO publicly stating quantum was '15-30 years away' at CES 2025, Google's Willow announcement (alongside IonQ and D-Wave commercial proof points) directly rebutted that timeline within days — reshaping market sentiment.

BEST PRACTICES

- ✓ Use Google's published benchmarks (13,000x speedup case) as a calibration point for what 'quantum advantage' actually looks like — narrow, verifiable, physics-simulation-shaped, not general AI.
- ✓ If pursuing Google Cloud + quantum, plan for a research-partnership engagement model rather than self-serve API access — Google's quantum access remains more curated than AWS Braket or IBM Quantum Network.
- ✓ Leverage TensorFlow Quantum if your team already has a TensorFlow-based ML stack — it's the lowest-friction entry point for hybrid QML experiments.
- ✓ Track Google's error-correction milestones as leading indicators for when fault-tolerant hardware becomes broadly available — their research often predates commercial availability by 2-3 years.

ANTI-PATTERNS TO AVOID

- ✗ Assuming Google's headline speedup numbers (13,000x) generalise to your workload — these are highly specific physics simulations, not a universal multiplier.
- ✗ Expecting the same level of self-service cloud access as AWS Braket or IBM Quantum — Google's commercial quantum offering is less mature for direct enterprise procurement.
- ✗ Building a quantum strategy around Cirq alone without Qiskit interoperability — most enterprise quantum talent and tooling has converged on Qiskit, creating hiring and ecosystem risk.
- ✗ Treating error-correction research announcements as immediately deployable capability — these are multi-year-ahead signals, not current-quarter features.

Sources: BQP Sim 'Quantum Computing & AI Together 2026'; Constellation Research 2025 Year in Review; SpinQ Global Footprint of Quantum Computers 2025.

SECTION 4

Microsoft Azure Quantum — Topological Qubits & Quantum Ready

Microsoft Azure Quantum

Quantum Ready Initiative • Topological qubit research bet

THE PROBLEM

Most enterprises have no internal framework to even ASSESS whether a given business problem is quantum-suitable, what hardware modality fits, or how to budget and skill up — the barrier isn't access to quantum hardware, it's organisational readiness.

THE SOLUTION / APPROACH

Microsoft's primary 2025-2026 quantum play was organisational rather than purely technical: the Quantum Ready Initiative helps enterprises systematically identify quantum use cases and build internal readiness BEFORE hardware capability fully matures — while simultaneously pursuing a long-shot but potentially category-defining bet on topological qubits (theoretically far more stable than superconducting or trapped-ion approaches).

KEY PRODUCTS & PLATFORMS

- Azure Quantum — multi-hardware cloud access (IonQ, Quantinuum, Rigetti via Azure)
- Quantum Ready Initiative — enterprise readiness assessment & roadmap programme
- Q# — quantum programming language integrated with Visual Studio / .NET tooling
- Topological qubit research programme (long-term differentiated hardware bet)
- Azure Quantum Elements — chemistry & materials simulation cloud service

FLAGSHIP USE CASES & RESULTS

Use Case		Domain	Result / Metric
Quantum assessment	readiness	Cross-industry	Quantum Ready Initiative launched specifically to help enterprises 'sort out use cases for quantum computing and get ready to deploy' (Constellation Research, 2025)
Chemistry simulation	& materials	Pharma / Manufacturing	Azure Quantum Elements targets materials discovery workflows integrated with Azure HPC
Multi-hardware development	algorithm	Cross-industry	Azure Quantum's multi-provider model (IonQ, Quantinuum, Rigetti) lets enterprises benchmark across modalities without separate vendor contracts

BIG WINS

Quantum Ready Initiative reframed the market conversation from 'is quantum real' to 'are YOU ready' — shifting competitive positioning toward organisational/consulting value rather than pure hardware specs.

Multi-hardware Azure Quantum marketplace lets Microsoft capture quantum cloud revenue regardless of which hardware modality ultimately wins — a portfolio hedge competitors with proprietary hardware can't match.

Deep .NET/Visual Studio integration via Q# lowers the barrier for the largest existing enterprise developer base to begin quantum experimentation inside familiar tooling.

BEST PRACTICES

- ✓ Start with a structured Quantum Ready-style readiness assessment BEFORE procuring any quantum compute — identify use cases, skills gaps, and data pipeline dependencies first.
- ✓ Use Azure Quantum's multi-hardware access to run the SAME algorithm across IonQ, Quantinuum, and Rigetti backends — empirically determine which modality fits your problem rather than betting on one vendor's roadmap.
- ✓ If your enterprise is .NET/Azure-centric, prototype in Q# to minimise tooling friction for your existing developer base.
- ✓ Treat the topological qubit programme as a multi-year watch item, not a current dependency — plan architectures that are hardware-agnostic so you can swap in topological qubits later without redesign.

ANTI-PATTERNS TO AVOID

- ✗ Skipping the readiness assessment and jumping straight to procurement — Microsoft's own positioning (and McKinsey's data showing budgets going to 'use-case development' over hardware) confirms readiness gaps are the dominant blocker.
- ✗ Building exclusively in Q# without Qiskit compatibility — limits portability if you later need to run on IBM or pure Qiskit-ecosystem partners.
- ✗ Waiting for topological qubits before starting any quantum work — this is a multi-year research bet; near-term value comes from superconducting/trapped-ion access available today via Azure Quantum.
- ✗ Treating Azure Quantum Elements as a drop-in replacement for existing computational chemistry tools without validating accuracy on your specific molecule classes first.

Sources: Constellation Research 'Quantum Computing Development Accelerates' 2025 Year in Review; McKinsey Quantum Technology Monitor 2026.

SECTION 5

AWS Braket — Multi-Provider Hybrid HPC Strategy

AWS Braket

Ocelot chip (cat qubits) • Quantum Embark Program

THE PROBLEM

Enterprises want to experiment with quantum computing without committing to a single hardware vendor, without standing up new infrastructure, and without disrupting existing classical HPC workflows that quantum must eventually integrate with.

THE SOLUTION / APPROACH

AWS Braket's strategy is pure platform aggregation plus pragmatic hybrid integration: offer every major hardware modality (superconducting, trapped-ion, neutral atom, annealing, photonic) through one pay-as-you-go API tightly woven into existing AWS HPC and ML services (SageMaker), while also investing in proprietary hardware (Ocelot, AWS's first chip, using novel 'cat qubit' architecture to reduce error-correction overhead).

KEY PRODUCTS & PLATFORMS

- Amazon Braket — unified API across Rigetti, IonQ, D-Wave, QuEra, Oxford Quantum Circuits, Xanadu, IQM
- Ocelot chip (2025) — AWS's first proprietary QPU, cat-qubit architecture for noise suppression
- Quantum Embark Program — structured enterprise adoption pathway (discovery -> algorithm dev -> deployment)
- Braket + SageMaker integration — hybrid quantum-classical ML pipelines
- Hybrid Jobs — managed execution combining classical optimisers with quantum circuit evaluation

FLAGSHIP USE CASES & RESULTS

Use Case	Domain	Result / Metric
Multi-hardware algorithm benchmarking	Cross-industry	Single API access to Rigetti (84-qubit Ankaa-2), IonQ, D-Wave annealers, QuEra neutral atoms — enterprises benchmark before committing
Quantum annealing for optimisation	Logistics / Manufacturing	D-Wave annealers deployed via AWS data centers, used by enterprises like Volkswagen Labs for routing/scheduling problems
Error-suppressed hardware R&D;	Hardware / Infrastructure	Ocelot's cat-qubit design (14 physical qubits incl. buffers/detection) targets dramatically reduced error-correction overhead vs standard superconducting approaches
Structured enterprise onboarding	Cross-industry	Quantum Embark Program provides modular use-case discovery through deployment — directly addressing the 'readiness gap' McKinsey identified as the top spending category

BIG WINS

Became the de facto 'quantum hardware app store' — enterprises can trial 6+ distinct hardware modalities without separate procurement relationships, dramatically lowering experimentation cost.

Ocelot (Feb 2025) demonstrated AWS isn't purely an aggregator — its cat-qubit architecture is a genuine hardware innovation specifically targeting the error-correction overhead problem that gates fault-tolerant computing.

AWS's HPC-first framing ('integrate quantum alongside traditional supercomputing' per PatentPC analysis) matches how McKinsey found enterprise budgets are actually being spent — on integration with existing tech stacks, not standalone quantum infrastructure.

BEST PRACTICES

- ✓ Use Braket's multi-provider access to run identical algorithms across 2-3 hardware modalities before any procurement commitment — empirical hardware selection beats vendor-roadmap-based selection.
- ✓ Enroll in the Quantum Embark Program rather than building a use-case discovery process from scratch — it's purpose-built for exactly the 'use-case development' spending category McKinsey identified as dominant.
- ✓ Architect quantum workloads as Hybrid Jobs from day one — even simulator-only prototypes should use the SageMaker + Braket Hybrid Jobs pattern so production migration requires no redesign.
- ✓ Treat D-Wave annealing access as the lowest-friction entry point for combinatorial optimisation problems (routing, scheduling) — it requires no circuit-level quantum programming expertise.

ANTI-PATTERNS TO AVOID

- ✗ Provider-shopping indefinitely without ever committing to a benchmark-and-decide cycle — Braket's breadth can become analysis paralysis if not time-boxed.
- ✗ Running quantum annealing (D-Wave) algorithms expecting gate-model quantum behaviour — annealers solve a different problem class (QUBO/Ising) and require different problem formulation.
- ✗ Building custom HPC-quantum integration glue code instead of using Braket Hybrid Jobs — reinvents infrastructure AWS already manages.
- ✗ Ignoring Ocelot and AWS's proprietary hardware roadmap while assuming AWS will remain 'just an aggregator' — AWS is increasingly a hardware vendor too, with its own roadmap dependencies.

Sources: *The Quantum Insider 'Top Quantum Computing Companies' (2026 update)*; PatentPC *'Quantum Cloud Computing: AWS, Google, IBM'*; Constellation Research *2025 Year in Review*.

SECTION 6

NVIDIA — CUDA-Q & GPU-Quantum Acceleration

NVIDIA

CUDA-Q platform • GPU-accelerated quantum simulation

THE PROBLEM

Quantum circuit simulation, variational algorithm training, and noise-model benchmarking all require massive classical compute — and quantum hardware itself is bottlenecked by classical control systems, error-correction decoders, and pre/post-processing that run on classical infrastructure alongside the QPU.

THE SOLUTION / APPROACH

NVIDIA positioned itself not as a quantum hardware vendor but as the essential classical compute layer underneath every quantum system — GPU-accelerated simulators (letting algorithm developers iterate on 40+ qubit circuits without QPU access), real-time error-correction decoders, and CUDA-Q as a unified programming model spanning CPUs, GPUs, and QPUs from multiple hardware partners.

KEY PRODUCTS & PLATFORMS

- CUDA-Q — open-source platform unifying GPU + QPU programming
- cuQuantum — GPU-accelerated state vector and tensor network simulators
- DGX Quantum — reference architecture pairing GPUs with QPU control systems for low-latency hybrid loops
- Partnerships with IonQ, Quantinuum, Rigetti, Quantum Brilliance, ORCA, and others for CUDA-Q integration

FLAGSHIP USE CASES & RESULTS

Use Case	Domain	Result / Metric
Large-scale circuit simulation	Algorithm Development	cuQuantum enables simulation of 40+ qubit circuits on GPU clusters, letting teams validate VQE/QAOA ansatze before consuming expensive QPU time
Real-time error correction	Hardware Infrastructure	DGX Quantum architecture targets the microsecond-scale classical feedback loops required for quantum error correction decoders
Cross-hardware algorithm portability	Cross-industry	CUDA-Q's single programming model lets teams write once and target IonQ, Quantinuum, Rigetti, or GPU simulators interchangeably

BIG WINS

Despite NVIDIA's own CEO publicly downplaying quantum timelines (15-30 years) at CES 2025, NVIDIA simultaneously built the indispensable classical infrastructure layer that EVERY quantum hardware vendor depends on — a hedge regardless of which hardware modality wins.

cuQuantum became a de facto standard for algorithm developers to prototype on GPUs before burning QPU time/budget — directly addressing the cost-of-experimentation barrier that slows enterprise quantum adoption.

By integrating with IonQ, Quantinuum, Rigetti AND Microsoft/AWS/Google's quantum stacks simultaneously, NVIDIA avoided picking a hardware-modality 'side' — a structurally stronger position than any single QPU vendor.

BEST PRACTICES

- ✓ Prototype and debug all variational algorithms (VQE, QAOA, QNN) on cuQuantum GPU simulators FIRST — only move to real QPU hardware once the algorithm converges reliably in simulation.
- ✓ Use CUDA-Q as your portability layer if you anticipate needing to switch QPU vendors — it decouples your algorithm code from any single hardware partner's SDK.
- ✓ For error-mitigation research requiring fast classical feedback loops, evaluate the DGX Quantum reference architecture rather than building custom low-latency classical-quantum interfaces.
- ✓ Budget GPU simulation time as a first-class line item — it is dramatically cheaper than QPU time and catches the majority of algorithm bugs before they consume quantum budget.

ANTI-PATTERNS TO AVOID

- ✗ Treating GPU simulation results as proof of quantum advantage — simulators run efficiently precisely BECAUSE the circuit is small/structured enough to be classically tractable; real advantage requires regimes simulators can't reach.
- ✗ Skipping GPU-simulator validation and debugging directly on QPU hardware — wastes expensive quantum compute time on bugs that are trivial to catch classically.
- ✗ Assuming CUDA-Q portability is 100% free — some hardware-specific optimisations (native gate sets, connectivity) still require backend-aware tuning even with a unified programming model.
- ✗ Over-indexing on NVIDIA's public skepticism about quantum timelines as a reason to deprioritise quantum investment — the same company is simultaneously building critical quantum infrastructure, a signal that's arguably more important than the soundbite.

Sources: Constellation Research 2025 Year in Review (NVIDIA CES 2025 commentary); cross-referenced with public CUDA-Q / cuQuantum documentation and partner announcements.

SECTION 7

Cross-Giant Patterns: What Separates Winners from Hype

Across all five giants, four structural patterns explain who is converting quantum investment into real enterprise traction versus who is generating press releases.

Pattern 1: Hybrid-First, Not Quantum-First

Every credible win (IBM's Project Bob inside SDLC, AWS Hybrid Jobs, Microsoft Azure Quantum Elements inside HPC pipelines) embeds quantum as ONE COMPONENT within an existing classical workflow. None of the winning patterns require ripping out existing infrastructure. Architect takeaway: never propose a 'quantum-first' redesign — propose a hybrid insertion point.

Pattern 2: Readiness Spend Dominates Hardware Spend

McKinsey's data is unambiguous: enterprise quantum budgets flow overwhelmingly to 'use-case and application development, integration with existing tech stacks, and internal capability building' — NOT hardware or OS development. Microsoft's Quantum Ready Initiative and AWS's Quantum Embark Program are direct responses to this. Architect takeaway: budget for readiness assessment and integration work as the LARGEST line items, not QPU access fees.

Pattern 3: Verifiable, Narrow Benchmarks Build Credibility

Google's 13,000x speedup claim was credible specifically because it was narrow (one physics simulation), verifiable, and immediately referenced by competitors. Vague 'quantum will transform your business' claims (common in 2023-2024 marketing) have given way to specific, falsifiable benchmarks. Architect takeaway: demand specific benchmark results on problem classes similar to yours before trusting any vendor's general claims.

Pattern 4: Multi-Hardware Hedging Is the Dominant Strategy

Microsoft (Azure Quantum), AWS (Braket), and NVIDIA (CUDA-Q) all explicitly avoid betting on one hardware modality, instead building abstraction layers that work across superconducting, trapped-ion, neutral atom, annealing, and photonic systems. Only IBM and Google maintain strong proprietary-hardware-first positioning — and even they offer Qiskit/Cirq portability. Architect takeaway: any architecture committing irreversibly to one QPU vendor's proprietary SDK without an abstraction layer is taking on unnecessary technology risk.

SECTION 8

Master Anti-Pattern Library

A consolidated list of the highest-frequency mistakes observed across tech giant case studies, organised by failure category.

Strategic Anti-Patterns

- ✗ Framing quantum as a 'transformation' rather than an 'augmentation' of existing systems — every credible 2025-2026 win is additive, not replacement.
- ✗ Committing budget to QPU access before completing a use-case readiness assessment — inverts the actual spending pattern of successful adopters (McKinsey).
- ✗ Selecting a hardware vendor based on roadmap promises rather than benchmarked results on your problem class.

Technical Anti-Patterns

- ✗ Skipping classical simulator validation (cuQuantum or equivalent) before consuming QPU budget on unproven algorithms.
- ✗ Hard-coding to one vendor's proprietary SDK (bypassing Qiskit/CUDA-Q portability layers) without an explicit, documented reason.
- ✗ Designing real-time/low-latency systems with quantum components in the critical path — current QPU latency (seconds to minutes including queue time) makes this infeasible.
- ✗ Using gate-model quantum algorithm patterns (VQE, QAOA circuit design) when the underlying problem is actually a QUBO/Ising formulation better suited to annealing hardware (and vice versa).

Organisational Anti-Patterns

- ✗ Running quantum pilots as isolated 'innovation lab' projects disconnected from production data pipelines and business stakeholders — the IBM Project Bob pattern (8,000 working devs, measurable productivity gain) succeeded specifically because it was embedded in real workflows from day one.
- ✗ Building in-house quantum chemistry/optimisation expertise from zero rather than leveraging the 280+ partner ecosystems (IBM Quantum Network) or structured onboarding programmes (Quantum Embark, Quantum Ready) that already encode this expertise.
- ✗ Treating a single press-release benchmark (e.g. '13,000x speedup') as evidence your specific workload will see similar gains — benchmarks are problem-class-specific, not universal multipliers.

SECTION 9

Sources & Further Reading

- IBM 3Q25 Earnings Prepared Remarks (SEC Form 8-K, October 2025) — [sec.gov](#)
- IBM Think — 'The Trends That Will Shape AI and Tech in 2026' (March 2026)
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- McKinsey — 'Quantum Technology Monitor 2026: A Commercial Tipping Point' (April 2026)
- LevelFields — 'Top Quantum Computing Stocks to Watch 2026' (April 2026)
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- PatentPC — 'Quantum Cloud Computing: How AWS, Google, and IBM Are Driving Adoption'
- BQP Sim — 'Quantum Computing & AI Together: 2026'
- SpinQ — 'The Surprising Global Footprint of Quantum Computers in 2025'
- TechAhead — 'The Role of Quantum Computing in Future LLMs' (February 2026)

This report is part of a 3-segment series. Segment 2 covers Consultancies (Accenture, Deloitte, McKinsey/QuantumBlack, BCG X) and Segment 3 covers Quantum-Native Startups (IonQ, Quantinuum, Xanadu, PsiQuantum, Multiverse Computing, D-Wave, Rigetti, and others).